

1 Classification Evaluation Example

#	Sentence	GT	CLA
1	The Limmat flows through the city	CH	CH
2	The Eiffel Tower is cool	FR	DE
3	The wall fell many years ago	DE	CH
4	Pilatus towers over the city	CH	FR
5	Nîmes has a Roman arena	FR	FR

$$F_{\text{CH}} = \frac{1}{2}, \quad F_{\text{FR}} = \frac{1}{2}, \quad F_{\text{DE}} = 0$$

$$F_{\text{macro}} = \frac{1}{3}$$

$$F_{\text{weighted}} = \frac{\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 2 + 0}{5} = 0.4$$

$$P_{\text{CH}} = \frac{1}{2}, \quad P_{\text{FR}} = \frac{1}{2}, \quad P_{\text{DE}} = 0$$

$$R_{\text{CH}} = \frac{1}{2}, \quad R_{\text{FR}} = \frac{1}{2}, \quad R_{\text{DE}} = 0$$

2 Classification Evaluation Example 2

#	Sentence	GT	CLA
1	The Limmat flows through the city	CH	CH
2	The Eiffel Tower is cool	FR	CH
3	The wall fell many years ago	DE	CH
4	Pilatus towers over the city	CH	FR
5	Nîmes has a Roman arena	FR	CH

$$F_{\text{CH}} = \frac{1}{3}, \quad F_{\text{FR}} = 0, \quad F_{\text{DE}} = 0$$

$$F_{\text{macro}} = \frac{1}{9}$$

$$F_{\text{weighted}} = \frac{\frac{1}{3} \cdot 2 + 0 \cdot 2 + 0 \cdot 1}{5} = \frac{2}{15}$$

$$P_{\text{CH}} = \frac{1}{4}, \quad P_{\text{FR}} = 0, \quad P_{\text{DE}} = 0/0$$

$$R_{\text{CH}} = \frac{1}{2}, \quad R_{\text{FR}} = 0, \quad R_{\text{DE}} = 0$$

Technically, since precision is undefined for DE, F_{DE} is also undefined.